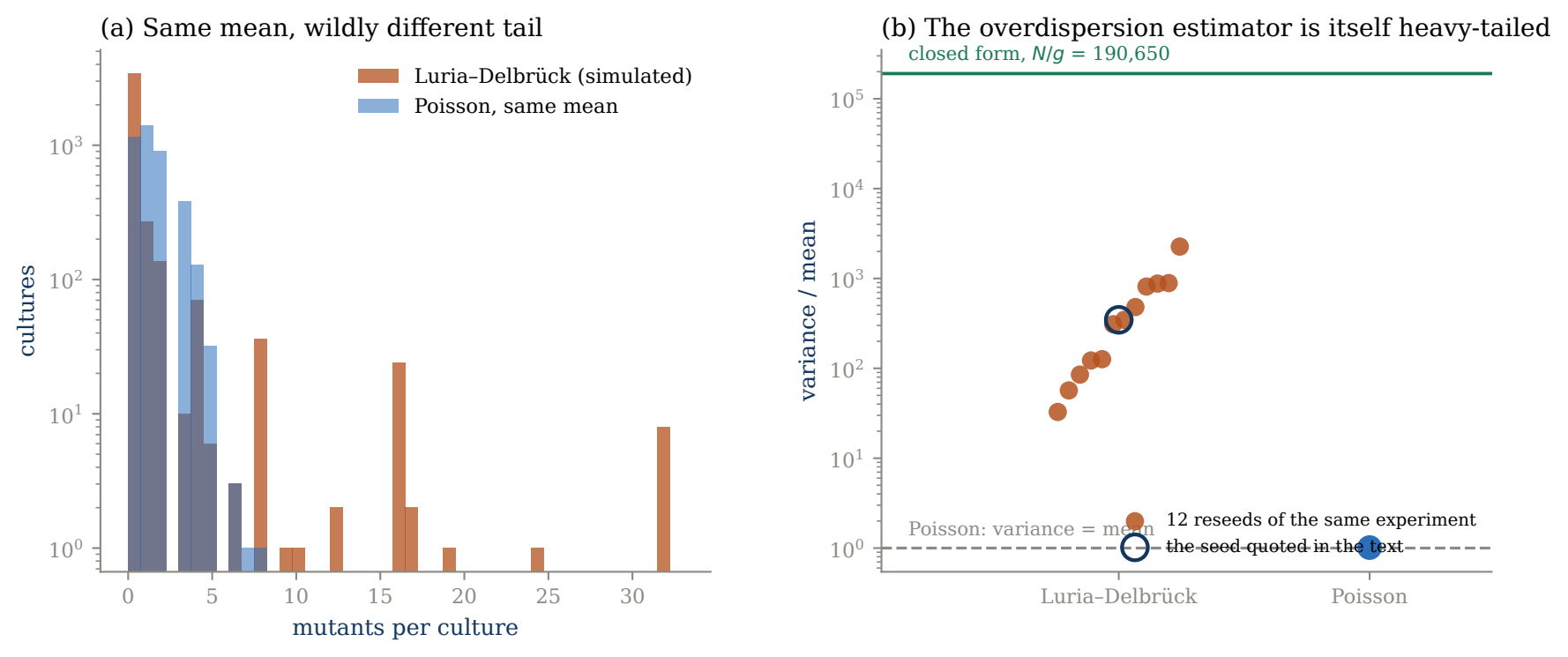
(a) Same mean, wildly different tail
Luria–Delbrück (simulated)
Poisson, same mean
cultures
10^3
10^2
10^1
10^0
0
5
10
15
20
25
30
mutants per culture
(b) The overdispersion estimator is itself heavy-tailed
closed form, N/g = 190,650
variance / mean
10^5
10^4
10^3
10^2
10^1
10^0
Poisson: variance = mean
12 reseeds of the same experiment
the seed quoted in the text
Luria–Delbrück
Poisson